

## Question

Write a Java program to demonstrate the inheritance relationship **C1 extends C2**. Use constructors, method overriding and method overloading to calculate Simple Interest.

## Steps

1. Create a parent class **Interest**.
2. Declare principal and time.
3. Create a constructor to initialize principal and time.
4. Define a **calculate()** method in the parent class.
5. Create a child class **SimpleInterest** extending **Interest**.
6. Declare rate of interest in the child class.
7. Create a constructor in the child class.
8. Use **super()** to call the parent constructor.
9. Override the **calculate()** method.
10. Overload the **calculate()** method with different parameters.
11. Create an object of the child class in **main()**.
12. Call the methods and display the Simple Interest.

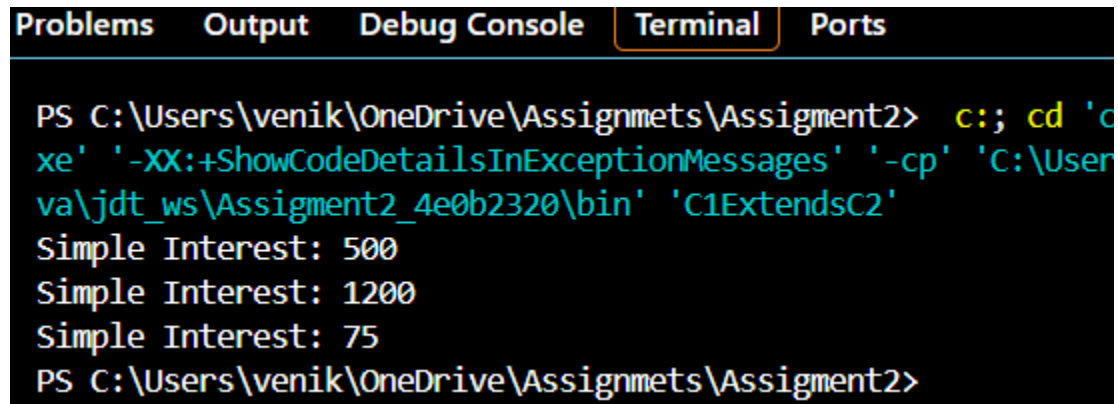
## Algorithm

START

1. Create class Interest.
2. Declare principal P and time T.
3. Initialize P and T using constructor.
4. Define calculate() method in Interest.
5. Create class SimpleInterest extending Interest.
6. Declare rate R.
7. Initialize R using constructor.
8. Call the parent constructor using super().
9. Override calculate().

10. Calculate  $SI = (P \times T \times R) / 100$ .
11. Overload calculate() with different parameters.
12. Create SimpleInterest object in main().
13. Call the calculate() methods.
14. Display the Simple Interest.

STOP



```
Problems  Output  Debug Console  Terminal  Ports

PS C:\Users\venik\OneDrive\Assignmets\Assignment2> c:; cd 'c:\Users\venik\OneDrive\Assignmets\Assignment2\bin' 'C1ExtendsC2'
Simple Interest: 500
Simple Interest: 1200
Simple Interest: 75
PS C:\Users\venik\OneDrive\Assignmets\Assignment2>
```